University of Chittagong

Department of Computer Science and Engineering

Assignment
Database Systems
Course Code: CSE 413

Submitted to,

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The overall goal in the whole mini-project is to create and work with a medical database in several incremental steps. The first step is to identify the information that is going to be represented as well as special requirements. Then, the information is modeled in an ER diagram and mapped to a relational schema. The database schema is then refined and normalized. Later on, database dumps will be provided and queried.

Preliminary Database Modeling

The database should store information about drugs, their categories, the diseases they can treat, their side effects, their commercial products, their interactions with other drugs, and possibly information about related clinical trials. In general, you may use the WebMD website (`https://www.webmd.com/`) for inspiration. But be aware that WebMD stores a large variety of information and that some information there is quite verbose. You should not try to incorporate all the information available in WebMD into your database design! For most aspects, it will be sufficient to have 5 attributes instead of 20. Please choose carefully.

Sample Information

To provide you with an idea of how the information to be loaded in the databasemight look like, consider the sample information available at the given excel file. Please notice thatthe sample information does not correspond to whatwould be a database table (or rela-tion).

Sample Queries

Your database model should allow for answering queries such as:

a) Find the number of drugs that have nausea as a side effect

b) Find the drugs that interact with butabarbital

c) Find the drugs with side effects cough and headache

d) Find the drugs that can be used to treat endocrine diseases

e) Find the most common treatment for immunological diseases that have not been used for hematological diseases

f) Find the diseases that can be treated with hydrocortisone but not with etanercept

Report

Please work in groups and solve the following tasks based on your current knowledge. You might want to make some assumptions about the mini-world, please clearly state them as part of your report.

1. Document and describe your database in the way you would do it (or have done it) in a project report.

2. Provide a list of tables with attributes that you would use to store your data.

3. In case you already have a good command of SQL, add the SQL commands that create your database.

Contents

Introduction

The objective of this mini-project is to design and document a database system that stores information about drugs, their categories, the diseases they treat, their side effects, and their interactions with other drugs. Additionally, it may contain information about related clinical trials. This project will follow a structured approach to database design, starting from an Entity-Relationship (ER) model to the final relational schema, ensuring data normalization and integrity.

Database Description

The database is designed to manage the following information:

- **Drugs**: Details of each drug, including its name, category, and general description.
- **Categories**: Different categories to which a drug may belong (e.g., analgesics, antibiotics).
- **Diseases**: Information about diseases, including their name and description.
- **Side Effects**: Potential side effects caused by each drug.
- **Interactions**: Information on how one drug may interact with another.
- **Clinical Trials**: Details of clinical trials related to specific drugs.

Assumptions

- Each drug belongs to one primary category.
- A drug can treat multiple diseases and have multiple side effects.
- Drug interactions are defined as pairs, where Drug A interacts with Drug B.
- For simplicity, clinical trials are associated directly with drugs without detailing individual study participants.

Relational Schema

- **Drug** (<u>Drug_ID</u>, Name, Category_ID, Description)
- **Category** (<u>Category_ID</u>, Name, Description)

- **Disease** (Disease_ID, Name, Description)
- **SideEffect** (SideEffect_ID, Description)
- **DrugSideEffect** (Drug_ID, SideEffect_ID)
- **DrugDisease** (Drug_ID, Disease_ID)
- **Interaction** (DrugA_ID, DrugB_ID, Description)
- **ClinicalTrial** (Trial_ID, Drug_ID, Description, Start_Date, End_Date)

SQL Commands

Below are the SQL commands to create the tables described in the relational schema:

```sql
CREATE TABLE Category (
Category_ID INT PRIMARY KEY,
Name VARCHAR(100),
Description TEXT
);

CREATE TABLE Drug (
Drug_ID INT PRIMARY KEY,
Name VARCHAR(100),
Category_ID INT,
Description TEXT,
FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID)
);

CREATE TABLE Disease (
Disease_ID INT PRIMARY KEY,
Name VARCHAR(100),
Description TEXT
);

CREATE TABLE SideEffect (
SideEffect_ID INT PRIMARY KEY,
Description TEXT
);

CREATE TABLE DrugSideEffect (
Drug_ID INT,
SideEffect_ID INT,
```

```
PRIMARY KEY (Drug_ID, SideEffect_ID),
FOREIGN KEY (Drug_ID) REFERENCES Drug(Drug_ID),
FOREIGN KEY (SideEffect_ID) REFERENCES SideEffect(SideEffect_ID)
);

CREATE TABLE DrugDisease (
Drug_ID INT,
Disease_ID INT,
PRIMARY KEY (Drug_ID, Disease_ID),
FOREIGN KEY (Drug_ID) REFERENCES Drug(Drug_ID),
FOREIGN KEY (Disease_ID) REFERENCES Disease(Disease_ID)
);

CREATE TABLE Interaction (
DrugA_ID INT,
DrugB_ID INT,
Description TEXT,
PRIMARY KEY (DrugA_ID, DrugB_ID),
FOREIGN KEY (DrugA_ID) REFERENCES Drug(Drug_ID),
FOREIGN KEY (DrugB_ID) REFERENCES Drug(Drug_ID)
);

CREATE TABLE ClinicalTrial (
Trial_ID INT PRIMARY KEY,
Drug_ID INT,
Description TEXT,
Start_Date DATE,
End_Date DATE,
FOREIGN KEY (Drug_ID) REFERENCES Drug(Drug_ID)
);
```

Conclusion

This database design is aimed at efficiently managing information about drugs, diseases, side effects, and clinical trials. It ensures normalization and avoids redundancy while allowing for complex queries related to drug interactions and treatment options for various diseases. This schema will serve as a basis for further refinement and analysis in subsequent project stages.